

STAT 24620 / FINM 34700 / STAT 32950

Group Data Project Instructions

Spring 2026

Due: May 22, 2026, 11:59pm

Purpose

The group project asks each group to apply several methods from the course to one real-world dataset chosen from the instructor-provided list. The goal is not to produce an exhaustive research paper. The goal is to compare what different multivariate and statistical learning methods reveal about the same dataset.

Dataset list

Dataset	File
Ames housing sales and property characteristics	ames_housing.csv
Capital Bikeshare hourly demand and weather	bike_sharing_hourly.csv
Italian air-quality sensor and reference measurements	air_quality_italy.csv
U.S. communities, socioeconomic variables, and violent crime	communities_crime.csv
NHANES adult health and examination measures	nhanes_health.csv

Required Components

Every group must complete the following core components. These are the basis for full credit.

1. Define a clear research question motivated by the dataset.
2. Conduct descriptive data exploration and visualization.
3. Apply at least two methods from the course to study structure, association, or prediction in the data.
4. At least one method should address multivariate structure, such as PCA, factor analysis, clustering, or mixture modeling.
5. At least one method should address prediction or supervised comparison, such as regression, logistic regression, LDA, ridge/lasso, trees, or random forests.
6. Compare the methods. Explain what each method answers, what it reveals, and how the conclusions differ.
7. Submit a clear, reproducible report with well-chosen figures or tables.

What Counts for Full Credit

A group can receive full credit without doing any extra analyses beyond the required components. Full credit depends on whether the core analyses are appropriate, correctly implemented, clearly interpreted,

and connected to the research question.

Report Requirements

Submit one group report, approximately 4–6 pages excluding code appendix. The report should include:

- research question and dataset description
- preprocessing decisions
- methods used and enough implementation detail to make the analysis reproducible
- 3–5 well-chosen figures or tables
- interpretation of the main results
- comparison across methods
- limitations
- full names of all group members

You should propose reasonable research questions (can be the same or similar questions as in the original papers).

Workload Guidance

These are real-world datasets, so some variables may be redundant, mixed-type, missing, transformed, or awkwardly coded. You are not expected to solve every data-cleaning issue. Make reasonable preprocessing choices and state them clearly.

AI Policy

The course AI policy applies. AI tools may be used for coding help, debugging, and grammar polishing, but not to generate explanations or complete answers. Any AI use should be briefly acknowledged.

Dataset Descriptions

1. Ames Housing

This dataset contains residential property sales in Ames, Iowa, with many structural, neighborhood, quality, and sale-condition variables. The **SalePrice** variable records the home sale price. The derived variable **HighSalePrice** indicates whether sale price is above the sample median.

Background. Housing prices are shaped by several overlapping sources of variation: lot size, building age, total living area, neighborhood, construction quality, garage and basement features, and market conditions at the time of sale. This dataset resembles a real appraisal or real-estate prediction problem where many predictors are correlated. For example, larger homes often have larger garages, more rooms, and higher quality ratings; PCA or clustering can help summarize these related features before prediction.

Source and original context. Ames housing data from De Cock (2011), “Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project,” *Journal of Statistics Education*. The CSV here was generated from the **AmesHousing** R package source archive. Source documentation: https://www.rdocumentation.org/packages/AmesHousing/versions/0.0.4/topics/ames_raw

Variables.

- Sales price: **SalePrice**; **HighSalePrice**, a derived binary label.
- Location and land: **MS_Zoning**, **Neighborhood**, **Lot_Area**, **Lot_Frontage**, **Lot_Shape**, **Land_Contour**, **Lot_Config**, **Land_Slope**.
- Building type and age: **Bldg_Type**, **House_Style**, **MS_SubClass**, **Year_Built**, **Year_Remod_Add**.
- Quality and condition: **Overall_Qual**, **Overall_Cond**, **Exter_Qual**, **Exter_Cond**, **Kitchen_Qual**, **Heating_QC**.
- Size and rooms: **Gr_Liv_Area**, **X1st_Flr_SF**, **X2nd_Flr_SF**, **Total_Bsmt_SF**, **Full_Bath**, **Half_Bath**, **Bedroom_AbvGr**, **Kitchen_AbvGr**, **TotRms_AbvGrd**.
- Basement and garage: **Bsmt_Qual**, **Bsmt_Cond**, **Bsmt_Exposure**, **BsmtFin_Type_1**, **Garage_Type**, **Garage_Finish**, **Garage_Cars**, **Garage_Area**.
- Sale variables: **Mo_Sold**, **Yr_Sold**, **Sale_Type**, **Sale_Condition**.
- Missing values: some missing values represent absent features, such as no alley, no pool, no fence, no fireplace, or no garage.

2. Capital Bikeshare Hourly Demand

This dataset contains hourly bike rental counts from the Capital Bikeshare system together with calendar, season, weather, temperature, humidity, and wind variables. The variable **cnt** is total hourly rental count. The derived variable **high_demand** indicates whether demand is above the sample median.

Background. Bike-sharing systems have strong demand cycles. Commuting hours, weekends, holidays, season, and weather can all change the number of rentals. The data are hourly, so you may expect strong time-of-day patterns and possible differences between casual riders and registered riders. This makes the dataset useful for comparing structure discovery, such as clustering hours into demand regimes, with supervised prediction of demand.

Source and original context. UCI Machine Learning Repository Bike Sharing Dataset, created from Capital Bikeshare system data with weather and seasonal information. Source documentation: <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

variables.

- Rental counts: `cnt`; `high_demand`, a derived binary label.
- Demand components: `casual`, `registered`, `cnt`. Since `cnt` equals `casual` plus `registered`, avoid using `casual` and `registered` as predictors when predicting `cnt`.
- Time and calendar: `dteday`, `season`, `yr`, `mnth`, `hr`, `holiday`, `weekday`, `workingday`.
- Weather: `weathersit`, `temp`, `atemp`, `hum`, `windspeed`.

3. Italian Air Quality

This dataset contains hourly air-quality measurements from an urban monitoring site in Italy. It includes responses from a gas multisensor device, reference pollution measurements from certified analyzers, and meteorological variables such as temperature and humidity. The project CSV has already been cleaned: the original -200 missing-value codes were handled, variables with heavy missingness were removed, rows without valid dates were removed, and remaining numeric missing values were median-imputed. The provided CSV contains no missing values.

Background. Air-quality monitoring often combines certified reference instruments with lower-cost sensors. Sensor readings are correlated with pollution levels, but they may also respond to humidity, temperature, and other environmental conditions. This dataset gives a realistic example where multivariate dependence, sensor redundancy, calibration, time-of-day patterns, and prediction are all relevant.

Source and original context. UCI Machine Learning Repository Air Quality Dataset. The UCI documentation describes hourly sensor responses and certified analyzer measurements from March 2004 to February 2005 in an Italian city. Source documentation: <https://archive.ics.uci.edu/dataset/360/air+quality>

variables. (see the source link for a full description of variables)

- Response options: `N02_GT`; `high_N02`, a derived binary label; or another reference concentration such as `CO_GT`, `C6H6_GT`, or `N0x_GT`.
- Reference concentration variables: `CO_GT`, `C6H6_GT`, `N0x_GT`, `N02_GT`.
- Sensor response variables: `PT08_S1_CO`, `PT08_S2_NMHC`, `PT08_S3_N0x`, `PT08_S4_N02`, `PT08_S5_O3`.
- Weather and time: `Date`, `Hour`, `T`, `RH`, `AH`.
- Note: the `GT` variables are reference measurements, while the `PT08` variables are sensor responses. Be explicit about whether reference measurements are predictors, responses, or validation variables.

4. Communities and Crime

This dataset combines U.S. community-level socioeconomic and law-enforcement variables with violent crime rates. Identifier columns were removed. Variables with more than 20% missingness were dropped, and remaining numeric missing values were median-imputed. The variable `ViolentCrimesPerPop` is normalized per-capita violent crime. The derived variable `HighViolentCrime` indicates whether the value is above the sample median.

Background. This is an observational community-level dataset, not an experiment. Many variables summarize population composition, income, education, family structure, housing, mobility, and urbanization. These variables are strongly correlated and historically sensitive, so the project should focus on association, prediction, and limitations rather than causal claims.

Source and original context. UCI Machine Learning Repository Communities and Crime Dataset. The UCI documentation describes data combined from the 1990 U.S. Census, the 1990 Law Enforcement Management and Administrative Statistics survey, and 1995 FBI Uniform Crime Reporting data. Source documentation: <https://archive.ics.uci.edu/dataset/183/communities+and+crime>

variables. (see the source link for a full description of variables)

- Per-capita violent crime: `ViolentCrimesPerPop`; `HighViolentCrime`, a derived binary label.
- Population and urbanization: `population`, `householdsize`, `numbUrban`, `pctUrban`, `LandArea`, `PopDens`.
- Race, ethnicity, and age composition: `racepctblack`, `racePctWhite`, `racePctAsian`, `racePctHisp`, `agePct12t21`, `agePct12t29`, `agePct16t24`, `agePct65up`.
- Income and poverty: `medIncome`, `medFamInc`, `perCapInc`, `NumUnderPov`, `PctPopUnderPov`, `pctWWage`, `pctWInvInc`, `pctWPubAsst`, `pctWRetire`.
- Education and employment: `PctLess9thGrade`, `PctNotHSGrad`, `PctBSorMore`, `PctUnemployed`, `PctEmploy`, `PctEmplManu`, `PctEmplProfServ`, `PctOccupManu`, `PctOccupMgmtProf`.
- Family and household structure: `MalePctDivorce`, `FemalePctDiv`, `TotalPctDiv`, `PctFam2Par`, `PctKids2Par`, `PctYoungKids2Par`, `PctTeen2Par`, `PersPerFam`, `PctWorkMom`.
- Immigration, language, and housing: `PctForeignBorn`, `PctImmigRecent`, `PctSpeakEnglOnly`, `PctNotSpeakEnglWell`, `PctPersOwnOccup`, `PctHousOwnOcc`, `PctHousOccup`, `MedRent`, `OwnOccMedVal`.
- Sensitive demographic variables are present. Avoid statements implying that group identity causes crime; focus on aggregate association, prediction, confounding, and measurement limitations.

5. NHANES Adult Health

This dataset contains adult health, demographic, examination, and lifestyle variables derived from the U.S. National Health and Nutrition Examination Survey. The file uses complete cases from selected variables and includes a derived binary variable `HighBP`, indicating systolic blood pressure (`BPSysAve`) at least 130 or diastolic blood pressure (`BPDiaAve`) at least 80.

Background. NHANES is a national survey program that combines interviews, physical examinations, and laboratory measurements. The data are useful for studying associations among demographics, body measurements, lifestyle, and health outcomes. This project version is a simplified adult complete-case subset, so it is easier to analyze than the full survey but still reflects real-world issues such as mixed variable types and observational interpretation.

Source and original context. NHANES R package, based on the U.S. National Center for Health Statistics National Health and Nutrition Examination Survey. Source documentation: <https://cran.r-project.org/web/packages/NHANES/NHANES.pdf>

variables. (see the source link for a full description of variables)

- Blood pressure measurements: `HighBP`; `BPSysAve`; `BPDiaAve`.

- Demographic and socioeconomic variables: Gender, Age, Race1, Education, MaritalStatus, HHIncome, Poverty, HomeOwn.
- Body and examination variables: Weight, Height, BMI, Pulse, BPSysAve, BPDiaAve, DirectChol, TotChol.
- Health and lifestyle variables: Diabetes, HealthGen, PhysActive, SmokeNow, AlcoholYear, SleepHrsNight.
- Notes: the file is a complete-case subset and does not use NHANES survey weights. Interpret results as an analysis of this sample, not as official U.S. population estimates.